

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

April 2013

CE216 Computer Organization & Peripherals

Date: 22.04.2013, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Show how the MRI and non-MRI tables can be stored in memory. [04]
(b) Write down the four way for selection of address for control memory. Explain any one of them in brief. [03]
- Q - 2 (a) Design digital circuit to control the function of Accumulator Register. [06]
(b) Draw and explain the block diagram of Computer hardware configuration. [04]
(c) Design a 4-bit combinational circuit decrements using four full-adder circuits. [04]

OR

- Q - 2 (a) The following register transfers are to be executed for the common bus system. For each transfer, specify (1) The binary value that must be applied to bus select inputs S2, S1 and S0. (2) The register whose LD control input must be active. (3) A memory read or write operation and (4) The operation in the adder and logic circuit. [06]
- a. $AR \leftarrow PC$
 - b. $IR \leftarrow M[AR]$
 - c. $M[AR] \leftarrow TR$
 - d. $AC \leftarrow DR, DR \leftarrow AC$
- (b) A computer has 16 registers, an ALU with 32 operations, and a shifter with eight operations, all connected to a common bus system. [04]
(1) Formulate a control word for a micro-operation.
(2) Show the bits of the control word that specify the micro-operations
 $R4 \leftarrow R5 + R6$
- (c) Draw the block diagram for the hardware that implements the following statements: [04]
 $x + yz : AR \leftarrow AR + BR$
where AR and BR are two n-bit registers and x, y and z are control variables. Include the logic gates for the control function.
- Q - 3 (a) Find Quotient Q and Remainder A for value $(656)_{10} = (1010010000)_2$ divided by $(29)_{10} = (11101)_2$. [08]
(b) Draw and explain flowchart for add and subtract operations of signed magnitude representation. [04]
(c) Write different ways of presenting Floating-point numbers. [02]

OR

- Q - 3 (a) Show the step-by-step multiplication process using Booth algorithm when the following binary numbers are multiplied. Assume 5-bit registers that hold signed numbers. The multiplicand in both cases is +15. [07]
- (1) $(+15) * (+13)$
 - (2) $(+15) * (-13)$
- (b) Explain followings: [04]
- (1) Control memory
 - (2) What are the basic assembler functions?
- (c) How fractional numbers are stored in 32-bit floating point representation. [03]

SECTION - II

- Q - 4 (a) Draw and explain design of signed magnitude addition and subtraction operations. [04]
- (b) Draw a space-time diagram for a six-segment pipeline showing the time it takes to process eight tasks. [03]
- Q - 5 (a) Answer the following: [06]
- (1) Differentiate between RISC and CISC.
 - (2) Give Flynn's Classification. Describe one of them with an example.
- (b) Convert the following arithmetic expressions from infix to reverse Polish notation. [04]
- (1) $A * B + C * D + E * F$
 - (2) $A * B + A * (B * D + C * E)$
 - (3) $A + B * [C * D + E * (F + G)]$
 - (4) $A * [B + C * (D + E)] / F * (G + H)$
- (c) What are the limitations on parallel execution? Explain in detail. [04]

OR

- Q - 5 (a) How to speed up procedure call and return using overlapped register window? Explain with neat diagram. [05]
- (b) Vectorizing compilers generally detect loops that can be executed on a pipelined vector computer. Are the vectorization algorithms used by vectorizing compilers suitable for MIMD machine parallelization? Justify your answer. [05]
- (c) The memory unit of a computer has 256K words of 32 bits each. The computer has an instruction format with four fields: an operation code field, a mode field to specify one of seven addressing modes, a registers address field to specify one of 60 processor registers, and a memory address. Specify the instruction format and the number of bits in each field if the instruction is in one memory word. [04]
- Q - 6 (a) Answer the following (Any Four) [12]
- (1) List and explain Basic computer instruction formats.
 - (2) What are the basic rules of the assembly language and how assembler generates the address symbol table?

- (3) What are the interrupts? Explain different types of interrupts.
- (4) A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.
- How many selection inputs are there in each multiplexer?
 - What size of multiplexer are needed?
 - How many multiplexers are there in the bus?
- (5) Perform the arithmetic operations below with binary numbers and with negative numbers in signed-2's complement representation. Use seven bits to accommodate each number together with its sign. In each case, determine if there is an overflow by checking the carries into and out of the sign bit position.
- $(+35) + (+40)$
 - $(-35) + (-40)$
 - $(-35) - (+40)$
- (b) Can the letter I be used as symbolic address in the assembly language program [02] defined for the basic computer? Justify your answer.
